

The Coordination Tax: Light-Speed Delay in a Self-Replicating Probe Swarm Bites Only When Hops Are Long

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Abstract—A swarm of self-replicating probes fills a stellar field star by star, each settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled; finite light-speed is its stated future work. We add that light-speed limit: a probe deciding at a star treats a distant star as settled only once the news, travelling at c , has arrived, so probes race for the same target from stale views and waste trips. Over a 32-seed paired ensemble on identical galaxies, the delay slows the time to fill the field by a median of about 0% for powered nearest-neighbour flight, about 30% for nearest-star slingshot flight, and about 50% for maximum-boost slingshot flight; every case still reaches 100%, so lag alone produces no permanent unsettled fraction. The lag scale is set by the dimensionless ratio of information-delay to travel-time per hop, which equals the probe speed in units of c ; whether it actually bites is decided by how non-local each hop is. Fast but local flight is nearly immune, because a probe that loses a race simply takes the star next door; the cost falls on long-range hops made from stale views. This refines the perfect-information picture: the slingshot speed-up is real, but under finite light-speed a meaningful fraction of it is eaten by uncoordinated long-range collisions.

Index Terms—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics

I. INTRODUCTION

A self-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. How long that takes is the exploration-timescale question, and the sharpest treatment of it for self-replicators is Nicholson and Forgan’s slingshot study [1]: probes that steal energy from the motion of the stars they pass (a gravitational slingshot, following the single-probe analysis of Forgan, Papadogiannakis and Kitching [2]) can explore roughly two orders of magnitude faster than probes under their own power.

That result rests on one convenient assumption, which its authors name as a limitation and leave for future work: every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another probe has just claimed, because it always knows. In a real galaxy no such oracle exists. News that a star has been settled can travel no faster than light, so a probe choosing its next target is acting on a picture of the field that is, for distant stars, years or centuries out of date.

This paper adds that light-speed limit to the model and measures what it costs. The question is not merely academic bookkeeping: it decides whether the celebrated slingshot speed-up survives contact with the finite speed of information, or whether a swarm racing outward on stale maps wastes much of the advantage colliding with itself. The answer, built from a controlled paired-ensemble experiment, is that the tax is real but sharply selective - it is set in scale by the probe’s speed and in bite by how far each hop reaches.

II. THE LIGHT-DELAYED BELIEF MODEL

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next target at star f in year Y believes a distant star i is settled only once that beacon’s light has had time to arrive:

$$\text{settled}(i) + \frac{\text{dist}(f, i)}{c} \leq Y. \quad (1)$$

Under the perfect-information assumption this gate collapses to “settled at any non-negative year,” and the model reproduces the earlier results bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target, they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival.

How much this matters is governed by a single dimensionless ratio: the information delay across a hop divided by the travel time across the same hop. For a hop of length d the delay is d/c and the travel time is d/v , so the ratio is

$$\Lambda = \frac{d/c}{d/v} = \frac{v}{c}, \quad (2)$$

the probe’s speed in units of c . A slow probe ($\Lambda \ll 1$) outpaces its own ignorance only trivially - by the time it crosses a hop, the news of that hop has long since propagated - so stale views barely arise. A fast probe carries a large Λ and routinely decides before the relevant news can reach it. At the powered cruise speed of the source model, $3 \times 10^{-5} c$, Λ is negligible; the effect can only appear once probes are fast, which is precisely the slingshot regime.

Two properties keep the experiment honest. First, knowledge is evaluated only at the decision star at the moment

of decision; news a probe's path happens to cross in mid-flight is ignored. This deliberately undercounts what a probe could know, so the measured waste is a conservative upper bound on the true penalty. Second, the whole simulation is a pure, seeded, fixed-step fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That is what makes a paired measurement of the delay's cost possible at all.

III. THE COORDINATION TAX

The cost was measured as a paired ensemble: 32 seeded galaxies of 300 stars each, every one filled twice - once with perfect information and once with the light-speed gate - so that the only difference between the two runs is the delay. Comparing the time to settle the entire field, the median slowdown from light-speed lag is:

- **powered nearest-neighbour flight: about 0%** - essentially immune;
- **nearest-star slingshot flight: about 30%** (interquartile range roughly +20% to +38%);
- **maximum-boost slingshot flight: about 50%** (interquartile range roughly +46% to +54%).

Two facts stand out. The first is that the penalty is *not* simply $\Lambda = v/c$. Powered flight is nearly untouched even though it is the slowest regime, and the two fast slingshot regimes differ from each other by nearly a factor of two. If Λ alone set the cost, the fastest probes would pay the most in a smooth progression; instead the cost tracks something else. That something is hop locality. A probe flying to the *nearest* star that loses a race has lost almost nothing: the next-nearest unsettled star is right beside it, so the wasted trip and the recovery are both short. A probe chasing the *biggest boost*, by contrast, deliberately reaches past its neighbours to a distant star; when it loses that race, both the wasted trip and the re-targeting detour are long. So $\Lambda \approx v/c$ sets the *scale* of the lag, but the *non-locality of each hop decides whether it bites*. Fast-but-local flight escapes; fast-and-far flight pays.

The second fact is what does *not* happen: every case still fills the field to 100%. Light-speed lag makes the swarm slower and more wasteful, but it never leaves a permanent unsettled remainder. Re-targeting guarantees that a probe which loses a race simply picks another star and eventually the field is closed. A permanent settled fraction below one - a steady state in which some region stays forever empty - requires settled sites to *die*, adding a loss term to the dynamics; delay by itself supplies no such term. Lag is a tax on speed, not a barrier to completion.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture. Nicholson and Forgan's speed-up is real and large [1], but it was computed under perfect coordination, and coordination is not free. Under finite light-speed a meaningful fraction of the advantage - roughly a third for nearest-star slingshots, half for maximum-boost - is spent on trips to stars that turn out

to be already taken. The more aggressively a policy reaches for distant, high-boost targets, the more of its speed it loses to uncoordinated collisions. There is a design lesson in that: for a real swarm on stale maps, a locally greedy policy is not merely simpler than a boost-maximising one, it is more robust to the delay it cannot avoid.

The behaviour is the expected signature of delayed multi-agent agreement. Formal consensus degrades as the ratio of communication delay to decision cadence grows: the standard protocol is stable only while the one-hop delay stays below a bound set by the coupling [3]. Our probes never reach agreement about the settled map at all - at a mean inter-star spacing of about a parsec the round-trip light-time is several years, far longer than any decision cadence - so they behave like agents forced to act on purely local, out-of-date views. This is the same regime that drove the space teleoperation and networking communities away from real-time control: past about a second of round-trip delay a human operator abandons closed-loop control for move-and-wait [4], and past minutes to hours no continuous end-to-end path exists and communication becomes hop-by-hop store-and-forward [5]. At interstellar hop lengths every one of those faster regimes is left far behind; each probe is effectively an independent colony acting on priors, and the coordination tax is what that independence costs when probes want the same stars.

The measurement is a conservative upper bound in two senses, and both point to future work. Probes here learn only at their decision stars, ignoring news their paths cross in flight, so a two-endpoint knowledge model would recover some of the wasted trips; and probes here share nothing directly, so adding probe-to-probe relay of the settled map would let a swarm partially coordinate rather than merely collide. Both would lower the tax without erasing it, since the fundamental limit - that no probe can know a distant star is taken until its light arrives - is set by physics, not by protocol. What this work establishes is the shape of the cost: bounded, never fatal to completion, scaled by v/c , and paid almost entirely by the long-range hops that made the swarm fast in the first place.

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